

Polynomial Regression: From First Principles

Linear Regression fits straight-line patterns, but many real relationships curve. Polynomial Regression extends Linear Regression by adding powers of the original feature so the model can learn curved patterns.

Why Polynomial Regression Is Needed

If study time improves exam score up to a point and then the benefit slows down, a straight line may not capture the pattern well. A curve can describe this kind of relationship better.

Core Formula

A simple polynomial model looks like this:

$$\hat{y} = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$

The model is still linear in the coefficients θ , but it is non-linear in the original input x .

Degree

The degree controls how flexible the curve is. Degree 1 is ordinary Linear Regression. Degree 2 creates a quadratic curve. Higher degrees create more flexible curves, but they can overfit.

How It Learns

The algorithm first creates polynomial features such as x^2 and x^3 . Then it fits Linear Regression on those transformed features.

Limitations

Polynomial Regression can overfit quickly when the degree is too high. Feature scaling and regularization are often important when using polynomial features.